

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0

- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Secure and Efficient Autonomous Systems Lab

May 2026 – Present

Undergraduate Research Assistant

East Lansing, MI

- Reproduced a published adversarial-robotics attack loop (attacker, target, judge, syntax-checker) as the baseline, then drove the decomposition attack to an 80% success rate against the per-step triage gate.
- Ran the node graph byte-identical in Gazebo and on a physical RoboMaster EP Core, driving the real robot over a custom C++ WiFi driver at a 30 Hz control loop with sub-20ms command latency.
- Stress-tested the security of an LLM-controlled robot, targeting the cheap per-step gate that screens its motion plans as the exact seam an attacker slips a harmful plan through, one innocent fragment at a time.

Robostangs (FRC Team 548)

Jun 2023 – May 2024

Software Captain

Northville, MI

- Anchored auto-aim on real-time, vision-corrected localization that fused two AprilTag cameras with swerve odometry on a dedicated 250Hz Java thread with latency compensation, holding pose accurate during aggressive motion.
- Built shoot-from-anywhere aiming that maps target distance to arm angle through a regression model, keeping candidate models swappable behind one interface so the team could A/B them against measured shot data.

PROJECTS

Dadei | *Python, pgvector, Deepgram, Whisper*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Trained a custom wake-word model to 90% recall at under one false accept per hour, gating the command pipeline so the assistant only opens the mic on demand instead of continuously streaming audio.
- Engineered a no-merge-first speaker-ID model using an EMA voice centroid plus a bounded prototype bank with floor-and-margin gating, abstaining into a new identity on ambiguity, attributing 96% of utterances to the correct speaker.
- Replaced single-signal context lookup with hybrid retrieval over pgvector that fuses vector similarity, lexical overlap, recency decay, and participant overlap, grounding each proposal under a fixed token budget.
- Split speech recognition into a Deepgram streaming path for live command latency and a batched Whisper path for ambient bulk transcription, keeping interactive commands responsive while bulk transcription stays cheap.

WizViz | *Pose Inference, Gesture Classification*

devpost.com/software/wizviz

Webcam fighting game driven by real-time body-gesture recognition, winner of the Interactive Media track at SpartaHack X

- Classified five distinct combat gestures from body landmarks with a scale-invariant geometric model normalized by torso length, making detection work across players and camera distances without any labeled training data.
- Ran pose inference asynchronously off the render loop with a monotonic timestamp guard that drops out-of-order results, sustaining 60 FPS with sub-20ms input lag instead of blocking on per-frame detection.

fliks | *Go, PostgreSQL*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Architected an ML rating service that grades each clip independently and shows its verdict beside the crowd's, with those crowdsourced ratings serving as the labeled data the classifier trains on.

TECHNICAL SKILLS

Languages Python, C++, Java, SQL, Bash

AI / ML PyTorch, Deepgram, Whisper

Robotics Gazebo, AprilTag

Databases PostgreSQL, pgvector

Tools Git, Linux